# Jerry (Jiarui) Zhang

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## Education

**University of Southern California** **August 2022–Present**
*Ph.D. in Computer Science, Advisor: Willie Neiswanger, GPA: 3.78/4.00* *Los Angeles, USA*

**Tsinghua University** **August 2018–June 2022**
*Bachelor of Science in Electrical Engineering, GPA: 3.64/4.00* *Beijing, China*

## Publications (* denotes equal contribution)

Google Scholar

- **J Zhang***, M Tao*, S Wang*, O Liu, X Ma, W Neiswanger. *'An Exam for Active Observers'*, arXiv 2026. [Paper] [Project]
- **J Zhang**, M Khayatkhoei, P Chhikara, F Ilievski. *'MLLMs Know Where to Look: Training-free Perception of Small Visual Details with Multimodal LLMs'*, ICLR 2025.
- **J Zhang**, O Liu, T Yu, J Hu, W Neiswanger. *'Euclid: Supercharging Multimodal LLMs with Synthetic High-Fidelity Visual Descriptions'*, In submission, 2025.
- **J Zhang***, J Hu*, M Khayatkhoei, F Ilievski, M Sun. *'Exploring Perceptual Limitations of Multimodal LLMs on Small Visual Objects'*, TMLR 2026.
- Y Jiang*, **J Zhang***, K Sun*, Z Sourati, K Ahrabian, K Ma, F Ilievski, J Pujara. *'MARVEL: Multidimensional Abstraction and Reasoning through Visual Evaluation and Learning'*, NeurIPS D&B Track 2024.
- **J Zhang**. *'Guided Profile Generation Improves Personalization with LLMs'*, EMNLP 2024.
- K Ahrabian, Z Sourati, K Sun, **J Zhang**, Y Jiang, F Morstatter, J Pujara. *'The curious case of nonverbal abstract reasoning with multi-modal large language models'*, COLM 2024.
- **J Zhang**, F Ilievski, K Ma, A Kollaa, J Francis, A Oltramari. *'A Study of Situational Reasoning for Traffic Understanding'*, KDD 2023.
- P Chhikara, **J Zhang**, F Ilievski, J Francis, K Ma. *'Knowledge-enhanced agents for interactive text games'*, K-CAP 2023. (Best Student Paper Award)
- **J Zhang**, F Ilievski, K Ma, J Francis, A Oltramari. *'A Study of Zero-shot Adaptation with Commonsense Knowledge'*, AKBC 2022.
- X Yi, **J Zhang**, W Li, X Wang, X Xie. *'Clickbait Detection via Contrastive Variational Modelling of Text and Label'*, IJCAI 2022.

## Experience

**University of Southern California** **August 2022–Present**
*Graduate Student Researcher* *Los Angeles, USA*

- Developed training-free methods that use the attention and gradient maps of multimodal LLMs (MLLMs) to significantly improve their perception of small visual details, raising the accuracy of widely used MLLMs across a wide range of visual question answering benchmarks without training.
- Designed multimodal benchmarks that expose capability gaps in leading MLLMs: ActiveVision for active observation across 17 tasks with human baselines, Geoperception for 2D geometric perception, and MARVEL for abstract visual reasoning across 770 real puzzles.
- Analyzed how object quality, size, distractors, and location limit MLLM perception through controlled intervention studies, contributing new evaluation protocols for future MLLMs; showed that attention maps locate the relevant region even when answers are incorrect.
- Multimodal post-training: sweeping experiments across visual encoders, LLMs, training data composition, and curricula on high-fidelity synthetic data; delivered Euclid, a model strong at visual perception that also generalizes on multimodal reasoning benchmarks.

**Google Cloud** **August 2026–Present**
*Research Intern* *Sunnyvale, USA*

- Evaluate how well multimodal coding agents generate visual artifacts and communicate with target audiences.
- Explore vision-coding co-evolution, using visual feedback to improve how agents generate visual artifacts.

**Google DeepMind** **May 2025–August 2025**
*Student Researcher* *Los Angeles, USA*

- Built challenging, diagnostic benchmarks for multimodal language and reasoning models.

**Amazon Alexa AI** **May 2023–August 2023**
*Applied Scientist Intern* *Bellevue, USA*

- Improved the efficacy and efficiency of LLM-based personalization through step-wise self-optimization.